

```
#include <OneWire.h>
```

```
#include <BlynkRpcClient.h>
```

```
#define BLYNK_TEMPLATE_ID "TMPL3m1d1pAyw"
```

```
#define BLYNK_TEMPLATE_NAME "Smart Sipper"
```

```
#define BLYNK_AUTH_TOKEN "zvUh6I_p5WZseAouC-zImUC1poE5ZqhJ"
```

```
#include <WiFi.h>
```

```
#include <BlynkSimpleEsp32.h>
```

```
#include <OneWire.h>
```

```
#include <DallasTemperature.h>
```

```
#include <Wire.h>
```

```
#include <Adafruit_GFX.h>
```

```
#include <Adafruit_SH110X.h>
```

```
#include <time.h>
```

```
// WiFi
```

```
char ssid[] = "Loki";
```

```
char pass[] = "123456789";
```

```
// DS18B20
```

```
#define ONE_WIRE_BUS 4
```

```
OneWire oneWire(ONE_WIRE_BUS);
```

```
DallasTemperature sensors(&oneWire);
```

```
// Relay
```

```
#define RELAY_PIN 5
```

```
// OLED
```

```
Adafruit_SH1106G display(128, 64, &Wire, -1);
```

```
float tempC = 0;
```

```
BlynkTimer timer;
```

```
bool wifiPrinted = false;
```

```
bool blynkPrinted = false;
```

```
bool startupDone = false;
```

```
// 🔥 STARTUP SCREEN (STRICT BLOCK)
```

```
void showStartup() {
```

```
    display.clearDisplay();
```

```
    display.setTextColor(SH110X_WHITE);
```

```
    // Border UI
```

```
    display.drawRect(0, 0, 128, 64, SH110X_WHITE);
```

```
    display.setTextSize(1);
```

```
    display.setCursor(18, 18);
```

```
    display.print("SMART");
```

```
    display.setCursor(18, 38);
```

```
    display.print("SIPPER");
```

```
    display.setTextSize(1);
```

```
    display.setCursor(18, 55);
```

```
    display.print("Have a good day");
```

```
    display.display();
```

```
delay(4000); // 🔥 full visible
```

```
display.clearDisplay();
```

```
display.display();
```

```
}
```

```
void showLoading() {
```

```
  for (int i = 0; i <= 100; i += 5) {
```

```
    display.clearDisplay();
```

```
    display.setTextColor(SH110X_WHITE);
```

```
    // Border
```

```
    display.drawRect(0, 0, 128, 64, SH110X_WHITE);
```

```
    // Title
```

```
    display.setTextSize(1);
```

```
    display.setCursor(30, 15);
```

```
    display.print("Loading...");
```

```
    // Progress bar outline
```

```
    display.drawRect(14, 32, 100, 10, SH110X_WHITE);
```

```
    // Fill bar
```

```
    int width = map(i, 0, 100, 0, 98);
```

```
    display.fillRect(15, 33, width, 8, SH110X_WHITE);
```

```
    // Percentage
```

```
    display.setCursor(50, 48);
```

```
    display.print(i);
```

```
    display.print("%");
```

```

display.display();

delay(80); // speed of animation
}

delay(300);

display.clearDisplay();
display.display();
}

// ♦ Relay
BLYNK_WRITE(V1) {
  int relayState = param.asInt();
  digitalWrite(RELAY_PIN, relayState ? LOW : HIGH);
}

// ♦ Status print
void checkInitialStatus() {
  if (!wifiPrinted && WiFi.status() == WL_CONNECTED) {
    Serial.println("WiFi Connected");
    wifiPrinted = true;
  }
  if (!blynkPrinted && Blynk.connected()) {
    Serial.println("Blynk Connected");
    blynkPrinted = true;
  }
}

// ♦ Temperature
void readTemperature() {

```

```
sensors.requestTemperatures();

tempC = sensors.getTempCByIndex(0);


if (tempC != DEVICE_DISCONNECTED_C) {
    Blynk.virtualWrite(V0, tempC);
}


// Serial Debug
Serial.print("Temp: ");
Serial.println(tempC);


Serial.print("WiFi: ");
Serial.println(WiFi.status() == WL_CONNECTED ? "OK" : "NO");


Serial.print("Blynk: ");
Serial.println(Blynk.connected() ? "OK" : "NO");


Serial.println("-----");
}


// 🔥 OLED UI
void updateOLED() {

    if (!startupDone) return; // 🔥 BLOCK UI until startup done


    display.clearDisplay();
    display.setTextColor(SH110X_WHITE);


    bool wifiOK = WiFi.status() == WL_CONNECTED;
```

```

// Border

display.drawRect(0, 0, 128, 64, SH110X_WHITE);


// WiFi Icon

int bars = wifiOK ? 4 : 1;
for (int i = 0; i < bars; i++) {
    display.fillRect(4 + i*3, 12 - (i+1)*2, 2, (i+1)*2, SH110X_WHITE);
}


// Battery

display.drawRect(110, 2, 14, 7, SH110X_WHITE);
display.fillRect(112, 4, 10, 3, SH110X_WHITE);


// Clock

display.setTextSize(1);
display.setCursor(45, 2);


struct tm timeinfo;
if (getLocalTime(&timeinfo)) {
    char t[6];
    sprintf(t, "%02d:%02d", timeinfo.tm_hour, timeinfo.tm_min);
    display.print(t);
}


// Temp

display.setCursor(5, 18);


if (tempC == DEVICE_DISCONNECTED_C) {
    display.print("Temp:Err");
} else {
    display.print("Temp:");
}

```

```
display.print(tempC, 1);
display.print("C");
}

// Status
display.setCursor(5, 30);
display.print("Status:");

if (tempC == DEVICE_DISCONNECTED_C) {
    display.print("No");
}
else if (tempC < 25) {
    display.print("COLD");
}
else if (tempC <= 30) {
    display.print("NORMAL");
}
else {
    display.print("HOT!");
}

// Left side
display.setCursor(5, 42);
display.print("WiFi:");
display.print(wifiOK ? "OK" : "Conn");

display.setCursor(5, 54);
display.print("IP:");
if (wifiOK) display.print(WiFi.localIP()[3]);
else display.print("No");
```

```
// Right side

display.setCursor(70, 30);

display.print("Heat:");

display.print(digitalRead(RELAY_PIN) == LOW ? "ON" : "OFF");


display.setCursor(70, 42);

display.print("Cool:");

display.print(tempC > 30 ? "ON" : "OFF");


display.setCursor(70, 54);

display.print("App:");

display.print(Blynk.connected() ? "OK" : "NO");


display.display();
}
```

```
// 🛠️ SETUP
```

```
void setup() {
```

```
    Serial.begin(115200);
```

```
    sensors.begin();
```

```
    pinMode(RELAY_PIN, OUTPUT);
```

```
    digitalWrite(RELAY_PIN, HIGH);
```

```
    Wire.begin(21, 22);
```

```
    display.begin(0x3C, true);
```

```
// 🔥 STEP 1 → SHOW STARTUP FIRST
```

```
showStartup(); // First screen
```



```
showLoading(); // 🔥 New animation
```

```
// 🔥 STEP 2 → AFTER THAT CONNECT WIFI
```

```
WiFi.begin(ssid, pass);
```

```
configTime(19800, 0, "pool.ntp.org");
```

```
Blynk.config(BLYNK_AUTH_TOKEN);
```

```
Blynk.connect();
```

```
// 🔥 STEP 3 → ENABLE UI
```

```
startupDone = true;
```

```
// Timers
```

```
timer.setInterval(2000L, checkInitialStatus);
```

```
timer.setInterval(2000L, readTemperature);
```

```
timer.setInterval(1000L, updateOLED);
```

```
}
```

```
// LOOP
```

```
void loop() {
```

```
  Blynk.run();
```

```
  timer.run();
```

```
}
```